

Marabūt's Theory of Gravity: Transcending the Singularity – Electron Void Gravitational Spheres (EVGS)

Resolution of the Infinite Density Paradox

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Abstract

For over a century, the standard cosmological model has been haunted by the “singularity”—a mathematical breakdown yielding infinite density and zero volume at the core of a collapsed star. In physics, infinities signal the fatal failure of a model, not a profound physical reality. This manuscript, the eighth in the *Marabūt's Theory of Gravity* series, applies the Electron Flow Model (EFM) to permanently resolve this paradox. By redefining gravity as hydrodynamic fluid pressure rather than geometric curvature, we replace the dimensionless singularity with the Electron Void Gravitational Sphere (EVGS). The EVGS is a finite, ultra-dense Core Negative Sink with a measurable volumetric boundary and maximum thermodynamic compression. By enforcing finite structural bounds through the unified mechanics of vacuum flux, this framework eliminates the infinite density paradox, restores absolute compatibility with quantum mechanics, and preserves all observed macroscopic gravitational phenomena.

1 Introduction: Mathematical Failure of General Relativity

Under Albert Einstein's General Relativity [einstein2012], the gravitational collapse of a supermassive star inevitably leads to a point of infinite spacetime curvature. The legacy density equation dictates that as the radius approaches zero, the density diverges to infinity:

$$\rho = \lim_{r \rightarrow 0} \frac{M}{\frac{4}{3}\pi r^3} \rightarrow \infty \quad (1)$$

This mathematical divergence is not a profound physical truth; it is a fundamental breakdown of the geometric model. Infinities in physics represent a boundary where a theory ceases to function, creating irresolvable paradoxes that force the suspension of quantum mechanics—an anomaly that even Einstein struggled to reconcile geometrically [norton2024].

Marabūt's Theory of Gravity rejects physical infinities entirely. Through the application of the **Electron Flow Model (EFM)**, gravity is not an abstract warping of an empty spacetime fabric; it is the tangible, mechanical pressure of vacuum electron flux rushing to fill a thermodynamic void. What classical physics mischaracterizes as a “black hole” is, in reality, a localized fluid-dynamic drain: an **Electron Void Gravitational Sphere (EVGS)**.

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2 The Electron Void Gravitational Sphere (EVGS)

A collapsing star under EFM does not form a dimensionless point. Instead, it compresses into a finite super-lattice Core Negative Sink (Mara0). The EVGS strips electrons faster than the ambient flux can replenish them, creating an absolute negative pressure region. The resulting inward cascade produces all observed gravitational effects through pure hydrodynamic drag.

3 Finite Volume and the Unified Master Equation

The EVGS possesses a strict physical radius R_{evgs} and finite effective density ρ_{eff} . Surface gravity at the boundary is governed by the Unified Master Equation:

$$g_{\text{evgs}} = (\alpha_G) \cdot \rho_{\text{eff}} \cdot R_{\text{evgs}} \cdot \tau(T) \quad (2)$$

where $\alpha_G \approx 2.792 \times 10^{-7}$ (density in kg/m^3 , R in km), and $\tau(T)$ is the thermodynamic impedance. The core cannot collapse to zero volume because the hyper-compressed atomic lattice exerts a physical back-pressure, maintaining finite structure.



Figure 1: Conceptual Comparison: General Relativity Singularity vs. Finite EVGS Core.

4 Empirical Validation: The 53-Object Main Data Matrix

Applying the EFM equations to 53 peer-reviewed celestial objects (ranging from stellar-mass to supermassive) yields strictly finite radii and densities in every case. For example:

Sagittarius A* (Mass $\approx 4.3 \times 10^6 M_{\odot}$):

- $R_{\text{evgs}} \approx 1.27 \times 10^7$ km
- $\rho_{\text{eff}} \approx 1.8 \times 10^{18}$ kg/m^3 (finite)
- This physical boundary directly replaces the legacy GR mathematical event horizon ($\approx 1.29 \times 10^7$ km) with a thermodynamic super-lattice.
- No mathematical divergence occurs.

The complete 53-object atlas [marabut2026m5] confirms that every object previously labeled a “black hole” is a physical, finite-volume EVGS engine. None produce infinite density.

5 Conclusion: The Dawn of Finite Astrophysics

The singularity is not a physical reality; it is the mathematical artifact of an incomplete, purely geometric model. By supplanting this divergence with the Electron Flow Model, we have successfully replaced impossible infinities with a mechanically sound, finite thermodynamic entity: the Electron Void Gravitational Sphere (EVGS).

This resolution does more than simply patch a theoretical hole in legacy physics; it fundamentally bridges the macroscopic universe with the quantum realm. An EVGS is governed by the exact same fluid-dynamic pressure gradients that dictate the behavior of subatomic particles, completely eliminating the need to suspend physical laws at an event horizon. The core cannot collapse to zero volume because the hyper-compressed atomic lattice exerts a physical back-pressure, maintaining absolute, finite structure.

With the singularity formally transcended, the EFM framework is now primed for large-scale applied kinematics. The subsequent manuscripts in this compendium will deploy this finite geometry to conquer the most stubborn anomalies in modern astrophysics. Manuscript 9 will utilize the rigid EVGS boundary to solve the Final Parsec Problem via Binary Cascade Superposition, and Manuscript 10 will unleash the *Marabūt EVGS Master Ledger*—an empirical, fluid-dynamic survey of 53 extreme celestial sinks confirming the Jet Ignition Threshold.

The era of infinite density is over. There is absolutely no physical requirement for an $r = 0$ singularity, as the universe is a finite, calculable fluid continuum governed by one undeniable truth: Gravity is Pressure, not curvature, and not an intrinsic mass property.

The Unified Fluid Cosmos Framework

This paper serves as **Manuscript 08 Version 1** in the comprehensive series, *Marabūt's Theory of Gravity*. It presents the **Electron Flow Model (EFM)**, a generative, fluid-dynamic framework designed to fundamentally replace General Relativity. The EFM shatters classical circularity with a single, undeniable foundational premise: **Gravity is Pressure, not curvature, and not an intrinsic mass property.**

Published Manuscripts in this Series:

- **Manuscript 01:**
The Push-Pull Mechanics of Vacuum Flux (The Foundational Framework)
Zenodo: <https://doi.org/10.5281/zenodo.20126552>
- **Manuscript 02:**
Resolving the Weyl Potential Anomaly
Zenodo: <https://doi.org/10.5281/zenodo.20127273>
- **Manuscript 03:**
Mechanistic Resolution of Jet Deflection in Cygnus X-1
Zenodo: <https://doi.org/10.5281/zenodo.20128198>
- **Manuscript 04:**
Quantum Mechanics of Stellar Chain Reactions and EVGS Formation
Zenodo: <https://doi.org/10.5281/zenodo.20070275>
- **Manuscript 05:**
The Heliospheric Grand Data Matrix
Zenodo: <https://doi.org/10.5281/zenodo.20128777>
- **Manuscript 06:**
Fluid-Dynamic Reinterpretation of Gravity, Black Holes, and Gravitational Waves
Zenodo: <https://doi.org/10.5281/zenodo.20129881>
- **Manuscript 07:**
Fluid-Dynamic Reinterpretation of the Kinematic Sunyaev-Zeldovich Effect
Zenodo: <https://doi.org/10.5281/zenodo.20130260>
- **Manuscript 08:**
Transcending the Singularity – Electron Void Gravitational Spheres (Current Paper)
Zenodo: <https://doi.org/10.5281/zenodo.20130802>

Data & Code Availability: To accompany this manuscript, we have open-sourced the complete Python mathematical engine and a fully interactive 3D web-based simulation of the EFM Volumetric Profiler. Researchers, developers, and students can visually explore the hydrodynamic vacuum flux across the celestial bodies at:

- **Interactive 3D EFM Volumetric Profiler:**
<https://marabutmarabut.github.io/Theory-of-Gravity>
- **Official GitHub Repository:**
<https://github.com/MarabutMarabut/Theory-of-Gravity>

Declarations

Author Contributions: C.B.M. and A.L.M. co-developed the theoretical framework. C.B.M. formalized the Unified Master Equation and computational matrices. A.L.M. conducted rigorous data validation and structural review. Both authors contributed equally to the final manuscript preparation.

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